

Restricted Planar Three-Body Problem, Convexity, and Global Surface of Section

**Presented to the MA in Mathematics Program in Partial Fulfillment of the
Requirements for Completion of the Program**



Katz School of Science and Health

Yeshiva University

New York, NY

July 2020

Yael Eisenberg

Advisor: Professor Marian Gidea, Mathematics

Contents

1	Introduction	2
2	Derivation of L_1	4
3	Setting up the PCRTBP	6
3.1	Equations	6
3.2	Levi Civita Transformation	8
4	Hessian Analysis	9
5	Theory on the Existence of Global Surface of Section for Convex Hamiltonians	12
6	Graphs	13
7	Conclusions	29
	Acknowledgements	30
	References	32

Abstract

The restricted planar three-body problem has many applications in celestial mechanics. In this paper, we discuss the three body problem in terms of the Sun, Earth, and a spacecraft. We analyze the Hamiltonian, Hessian, and energy curve, and generate graphs. Using the Sylvester criterion, we examine where convexity applies. This gives us information about the energy level necessary to assure the existence of a global surface of section.

1 Introduction

The planar three body problem considers three bodies on the same plane that are in motion under mutual gravity. Here we will consider the Sun, Earth, and a spacecraft. Our three body problem is restricted, which means that the third mass, the spacecraft, is small enough that it does not affect the orbits of the Sun or of Earth. Lastly, the Sun and Earth are assumed to have circular orbits, making this problem the planar circular restricted three body problem, or PCRTBP in short.

We will consider a coordinate system that rotates together with the Sun and Earth, so that the Sun and Earth are on the same axis, and the origin of the coordinate system is at their center of mass. The mass of the Sun is approximately 333,000 times the mass of Earth, therefore the center of mass on the Sun-Earth axis will be very close to the center of the Sun.

We will study the orbit of the spacecraft relative to the rotating coordinate system described above. This orbit is given by a system of differential equations. There are five Lagrange (equilibrium) points for the system of differential equations. Lagrange points are points in space where the combined gravitational force from surrounding bodies balances in such a way that another body (such as a spacecraft or a trojan) placed at a Lagrange point will maintain its position relative to the other bodies. In other words, a Lagrange point is an equilibrium point relative to the rotating coordinate system, i.e., a point where the velocity and acceleration equal zero: $\ddot{x} = \dot{x} = 0$, $\ddot{y} = \dot{y} = 0$.

We are dealing with a planar system, therefore $z = 0$.

The Lagrange points L_1 , L_2 , and L_3 are co-linear with the Sun and Earth, and are unstable center-saddle points. Therefore, if a body at L_1 , L_2 or L_3 is pushed slightly out of place, it will easily fall out of the equilibrium point. The L_4 and L_5 Lagrange points are not co-linear with the Sun and Earth, and the two triangles L_4 -Sun-Earth and L_5 -Sun-Earth are equilateral. L_4 and L_5 are linearly stable in our case, because the mass of the first body (the Sun) is at least 25 times the mass of the second body (Earth). If a body located at L_4 or L_5 is pushed slightly, it will slowly oscillate around that point. Due to the stability of L_4 and L_5 , planets tend to accumulate asteroids at these points; these are called trojans. Lagrange points are utilized by NASA to place satellites in strategic locations in space.

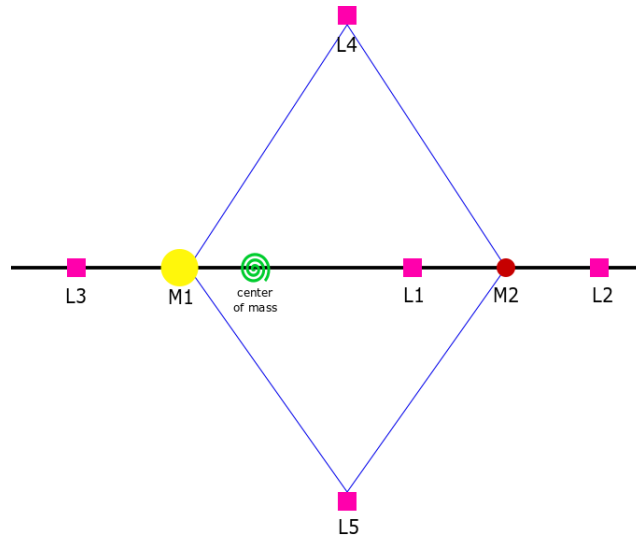


Figure 1: Lagrange points diagram: entire diagram rotates around the green center of mass

In figure 2, we notice the counterclockwise (direct) orbit around the Sun (M_1), and we see Earth (M_2) in relation to the Lagrange points. We can see how L_1 , L_2 , and L_3 are saddle unstable points, while L_4 , L_5 are stable points. In this figure, M_1 is the center of mass, since the center of mass on the Sun-Earth axis is very close to the center of the Sun (M_1). Since the diagram rotates along the route of the red counterclockwise orbit, the Lagrange points L_1, L_2, L_3, L_4, L_5 will orbit the Sun (M_1) with the same exact orbital period as Earth (M_2).

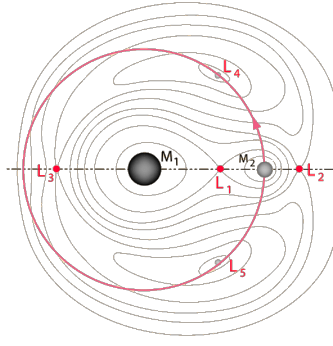


Figure 2: Orbits and Lagrange points (picture taken from Google images)

Additional background on the restricted three body problem and Hamiltonian systems can be found in [1], [2], [3], and background on celestial mechanics and dynamical systems can be found in [4].

2 Derivation of L_1

We will derive the approximate distance between L_1 and Earth, noting that the distance from Earth to L_2 will be similar, as L_1 and L_2 are approximately equidistant from Earth. Recall that the velocity of an object going on an orbit with respect to the radius AU and the period T is $V = \frac{2\pi AU}{T}$.

Assume we know the following values:

- 1 AU (or AU) = the astronomical unit, which is the mean distance from center of the Sun (center of mass) to the center of Earth. This is the radius of the orbit of Earth. 1 AU is approximately 149,597,871 km.
- M_1 = mass of the Sun
- M_2 = mass of Earth
- M_3 = mass of the spacecraft at Lagrange point L_1
- The distance from L_3 to the Sun is approximately $1 AU \times (1 + \frac{5M_2}{12M_1})$

- The distance from Earth to L_4 and L_5 is $1 AU$, at a 60° angle from the Sun-Earth axis. This happens because Sun-Earth- L_4 are equidistant from each other and Sun-Earth- L_5 are equidistant from each other as well, creating two equilateral triangles (see figure 1).

Firstly, we find all the forces acting on Earth: the spacecraft (located at L_1) and the Sun. As the spacecraft has negligible mass in comparison to the Sun, we can neglect the force on Earth due to the spacecraft. Therefore, we can assume that the only force acting on Earth is the gravitational force of the Sun. Recall that centripetal acceleration is velocity squared divided by the radius.

$$\Sigma F_E = M_2 a = \frac{M_2 V_E^2}{AU} = \frac{M_2 4\pi^2 AU}{T^2} \text{ by Newton's second law of motion.}$$

$$\Sigma F_E = \frac{GM_2 M_1}{AU^2} \text{ by Newton's law of universal gravitation.}$$

$$\text{Therefore, } \frac{4\pi^2 AU}{T^2} = \frac{GM_1}{AU^2}.$$

Secondly, we find the forces on the spacecraft due to the Sun and Earth. We choose to make the force created by the Sun positive, and the force created by Earth negative. This is because the axis we have set up has the spacecraft initially going towards the Sun (and away from Earth) on the x axis. Define $AU - L$ as the distance between the Sun and the spacecraft, and L as the distance from the spacecraft to Earth.

$$\Sigma F_L = M_3 a = \frac{M_3 V_L^2}{(AU-L)} = \frac{M_3 4\pi^2 (AU-L)^2}{(AU-L)T^2} \text{ by Newton's second law of motion.}$$

$$\Sigma F_L = GM_3 \left(\frac{M_1}{(AU-L)^2} - \frac{M_2}{L^2} \right) \text{ by Newton's law of universal gravitation.}$$

Putting this together:

$$GM_3 \left(\frac{M_1}{(AU-L)^2} - \frac{M_2}{L^2} \right) = \frac{M_3 4\pi^2 (AU-L)^2}{(AU-L)T^2} \rightarrow \frac{GM_1}{(AU-L)^2} - \frac{GM_2}{L^2} = \frac{4\pi^2 (AU-L)}{T^2} \rightarrow \frac{GM_1}{AU^2} \left(1 - \frac{L}{AU} \right)^{-2} - \frac{GM_2}{L^2} = \frac{4\pi^2 AU}{T^2} \left(1 - \frac{L}{AU} \right)$$

We use binomial approximation to simplify $\left(1 - \frac{L}{AU} \right)^{-2}$: $\left(1 - \frac{L}{AU} \right)^{-2} \approx \left(1 + (-2) \left(-\frac{L}{AU} \right) \right) = \left(1 + 2\frac{L}{AU} \right)$.

Earlier calculations indicate that $\frac{4\pi^2 AU}{T^2} = \frac{GM_1}{AU^2}$ since the period of Earth and the period of the spacecraft are identical.

It follows that:

$$\frac{GM_1}{AU^2} \left(1 + 2\frac{L}{AU} \right) - \frac{GM_2}{L^2} = \frac{4\pi^2 AU}{T^2} \left(1 - \frac{L}{AU} \right) = \frac{GM_1}{AU^2} \left(1 - \frac{L}{AU} \right) \rightarrow \frac{2M_1 L}{AU^3} - \frac{M_2}{L^2} = -\frac{M_1 L}{AU^3} \rightarrow L^3 = \frac{M_2 AU^3}{3M_1} \rightarrow L = \sqrt[3]{\frac{M_2}{3M_1}} AU$$

Hence, the distance from Earth to L_1 and from Earth to L_2 is approximately $\sqrt[3]{\frac{M_2}{3M_1}} AU$.

Using the values for M_1, M_2, AU given by NASA's website (and other resources), we use matlab to calculate the distance from Earth to L_1 to be approximately 1,496,396.43114536 kilometers. This is consistent with the L_1 data on NASA's website.

The above derivation gives the approximate distance from Earth to L_1 , as we used a binomial approximation. We can derive the exact location of L_1 by solving the 5th degree equation, which will give us an exact solution[5]:

$$p(x) = x^5 - (2+r)x^4 + (1+2r)x^3 - (1-r)x^2 + 2(1-r)x + r - 1 = 0, \text{ where } r = \frac{M_E}{M_S+M_E} \simeq 3.04042 \times (10^{-6}).$$

The x -coordinate of L_1 , where the Sun is located at the origin and Earth is located in position 1, will be the unique root of $p(x)$, in AU units. Matlab informs us that only real root is 0.989989026491005 which is the x coordinate of L_1 . Therefore, the distance from Earth to L_1 is $1 - 0.989989026491005 = 0.0100109735 AU$. We convert this value from AU units to kilometers to find that the distance from Earth to L_1 is approximately 1,497,620.32358306 km. When we compare this result with the approximated result calculated above, we find a very small error of approximately 0.08%, which verifies our result.

Next, we derive L_2 . The value of L_2 will be the only real root of $p(x) - 2rx^2 = 0$, in AU units. Matlab tells us that the only real root is 1.01007823667377, which is consistent with the diagram. We subtract 1 (location of Earth in AU units) from this value and convert the distance to kilometers. The result is that the distance from Earth to L_2 is approximately 1,507,682.74983011 km. The distance from Earth to L_2 is greater than the distance from Earth to L_1 , which is related to L_1 's location being much closer to the Sun in comparison with L_2 .

3 Setting up the PCRTBP

We will use the diagram in figure 1 to guide us, placing Earth at the origin of our coordinate system.

3.1 Equations

First, let us define the following parameters:

- $p = (p_1, p_2)$ is the velocity (linear momentum).

- $q = (q_1, q_2)$ is the position.

$$\langle p, q \rangle = p_1 q_1 + p_2 q_2, \text{ therefore } \langle p, iq \rangle = p_2 q_1 - p_1 q_2.$$

Note that the space (q_1, q_2, p_1, p_2) is 4-dimensional. For complex vectors $v = v_1 + iv_2$, $w = w_1 + iw_2$, we have

$$\langle v, w \rangle = \text{Re}(v \cdot \bar{w}) = \text{Re}[(v_1 + iv_2)(w_1 - iw_2)] = \text{Re}[v_1 w_1 + v_2 w_2 + i(w_1 v_2 - v_1 w_2)] = v_1 w_1 + v_2 w_2 \text{ which is the standard dot product of } (v_1, v_2) \cdot (w_1, w_2).$$

- M_1 is the mass of the Sun.
- M_2 is the mass of Earth.
- $\mu \in [0, 1]$ is the mass ratio $\mu = \frac{M_1}{M_1 + M_2}$. The approximate value for μ in the Sun-Earth system is $\mu \approx 0.999997$, however we will use various values for μ in the analysis.

The Hamiltonian we have to express the system (with all real values) $H : \mathbb{C} \setminus \{0, 1\} \times \mathbb{C} \rightarrow \mathbb{R}$ is given by:

$$H(q, p) = \frac{1}{2}|p|^2 + \langle p, iq \rangle - \langle p, i\mu \rangle - \frac{1-\mu}{|q|} - \frac{\mu}{|q-1|} \quad (1)$$

For this Hamiltonian, Earth is located at $(0,0)$ and the Sun is at $(1,0)$, unlike in the previous section where the center of mass (the Sun) is located at the origin.

- The equations of motion for the infinitesimal body are given by the Hamilton equations:
$$\begin{cases} \dot{q} = \frac{\partial H}{\partial p} \\ \dot{p} = -\frac{\partial H}{\partial q} \end{cases}$$

This system of differential equations determine the Hamiltonian flow $\phi^t(p_0, q_0) = (p(t), q(t))$ where $p(t), q(t)$ is the solution of the above Hamiltonian equations, with initial conditions $p(0) = p_0$, and $q(0) = q_0$.

- For $\mu \in \{0, 1\}$: H has a critical manifold diffeomorphic to the unit circle S^1 .
- For $\mu \notin \{0, 1\}$: H has the five critical/Lagrangian points, ordered by increasing the value of H .

As μ approaches 0 or 1 the critical value $H(L_1) \rightarrow -\frac{3}{2}$.

For $\mu = \frac{1}{2}$, we examine graphs for c where $-c = -H(L_1)$, where $L_1 = \frac{1}{2}$. Take $q = \frac{1}{2}$ and $p = 0$, and we plug the Hamiltonian in equation (1), giving us $H(\frac{1}{2}, 0) = -2$. Therefore, the second part of the analysis will be for $\mu = \frac{1}{2}$, and $-c$ being slightly less than -2 : $c = 2.01$.

In general, $-c$ represents the energy level. When $-c < H(L_1)$, the energy hypersurface $H^{-1}(-c)$ will have three connected components: one near earth, the second near the Sun, and the third near infinity. We will focus on the component closest to earth.

3.2 Levi Civita Transformation

Our Hamiltonian is $H(q, p) = \frac{1}{2}|p|^2 + \langle p, iq \rangle - \langle p, i\mu \rangle - \frac{1-\mu}{|q|} - \frac{\mu}{|q-1|}$ [6]. In these coordinates, the Hamiltonian system is not defined at $q = 0$ or $q = 1$ so $q = 0$ and $q = 1$ are singularities. Additionally, when q tends to 0, the velocity p approaches infinity (this situation corresponds to a collision between the spacecraft and Earth). For this reason, we create a coordinate transformation called the Levi Civita transformation to remove this singularity.

Define the following Levi-Civita coordinates [which defines a 2 : 1 map, symplectic up to a factor of 4]:

- $p = \frac{u}{v}$
- $q = 2v^2$

Therefore:

$$\begin{aligned}
 H(q, p) &= \frac{1}{2}|p|^2 + \langle p, iq \rangle - \langle p, i\mu \rangle - \frac{1-\mu}{|q|} - \frac{\mu}{|q-1|} \\
 &= \frac{1}{2}\left|\frac{u}{v}\right|^2 + \left\langle \frac{u}{v}, 2iv^2 \right\rangle - \left\langle \frac{u}{v}, i\mu \right\rangle - \frac{1-\mu}{|2v^2|} - \frac{\mu}{|2v^2-1|} \\
 &= \frac{1}{2}\frac{|u|^2}{|v|^2} + \operatorname{Re}\left(\left(\frac{u}{v}\right)(\overline{2iv^2})\right) - \operatorname{Re}\left(\left(\frac{u}{v}\right)(\overline{i\mu})\right) - \frac{1-\mu}{2|v|^2} - \frac{\mu}{|2v^2-1|} \\
 &= \frac{1}{2}\frac{|u|^2}{|v|^2} + 2\langle u, iv \rangle - \frac{\mu}{|v|^2} \operatorname{Im}(uv) - \frac{1-\mu}{2|v|^2} - \frac{\mu}{|2v^2-1|}
 \end{aligned}$$

Side calculations:

$$\operatorname{Re}\left(\left(\frac{u}{v}\right)(\overline{2iv^2})\right) = \operatorname{Re}\left(\left(\frac{u}{v}\right)(-2i\bar{v}^2)\right) = \operatorname{Re}(-2iu\bar{v}) = 2\langle u, iv \rangle$$

$$\operatorname{Re}\left(\left(\frac{u}{v}\right)(\overline{i\mu})\right) = \operatorname{Re}\left(\left(\frac{uv}{|v|^2}\right)(-i\mu)\right) = \frac{\mu}{|v|^2} \operatorname{Re}(-iuv) = \frac{\mu}{|v|^2} \operatorname{Im}(uv)$$

Following this, the Hamiltonian in the (v, u) coordinates is given by:

$$K_{\mu,c}(v, u) := |v|^2(H(q, p) + c) = \frac{1}{2}|u|^2 + 2|v|^2\langle u, iv \rangle - \mu \operatorname{Im}(uv) - \frac{1-\mu}{2} - \frac{\mu|v|^2}{|2v^2-1|} + c|v|^2 \quad (2)$$

We see that the energy level $-c$ for H corresponds to energy level 0 for K , and we have $\operatorname{Re}(dq \wedge d\bar{p}) = 4 \operatorname{Re}(dv \wedge d\bar{u})$.

Note that v is position and u is velocity in the new coordinates. However, we don't have collisions, since there is no $|v|$ in the denominator ($q = 0 \Leftrightarrow v = 0$). We can see that $H(v, u) = -c \Leftrightarrow K = 0$.

- For a fixed mass ratio $\mu \notin \{0, 1\}$: the component of the energy hypersurface $H^{-1}(-c)$ around Earth lifts to a compact component $\Sigma_{\mu,c}$ of the energy hypersurface $K_{\mu,c}^{-1}(0)$.
- The energy surface $K_{\mu,c}^{-1}(0)$ is diffeomorphic to S^3 .

4 Hessian Analysis

We compute the Hessian of (2):

$$K_{\mu,c}(v, u) := |v|^2(H(q, p) + c) = \frac{1}{2}|u|^2 + 2|v|^2\langle u, iv \rangle - \mu \operatorname{Im}(uv) - \frac{1-\mu}{2} - \frac{\mu|v|^2}{|2v^2-1|} + c|v|^2$$

The Hessian at (u, v) is described as a quadratic form in variables $(\hat{u}, \hat{v})$:

$$\begin{aligned}
D^2 K_{\mu, c}(u, v)[(\hat{u}, \hat{v}), (\hat{u}, \hat{v})] = & |\hat{u}|^2 + 2c|\hat{v}|^2 + 4\langle u, iv \rangle |\hat{v}|^2 + 8\langle v, \hat{v} \rangle \langle u, i\hat{v} \rangle \\
& + 8\langle v, \hat{v} \rangle \langle \hat{u}, iv \rangle + 4|v|^2 \langle \hat{u}, i\hat{v} \rangle \\
& - 2\mu \operatorname{Im}(\hat{u}\hat{v}) \\
& - \frac{48\mu|v|^2}{|2v^2 - 1|^5} (2|v|^2 \langle v, \hat{v} \rangle - \langle \bar{v}, \hat{v} \rangle)^2 \\
& + \frac{4\mu|v|^2}{|2v^2 - 1|^3} (4\langle v, \hat{v} \rangle^2 + 2|v|^2 |\hat{v}|^2 - \operatorname{Re}(\hat{v}^2)) \\
& + \frac{16\mu \langle v, \hat{v} \rangle}{|2v^2 - 1|^3} (2|v|^2 \langle v, \hat{v} \rangle - \langle \bar{v}, \hat{v} \rangle) \\
& - \frac{2\mu |\hat{v}|^2}{|2v^2 - 1|}
\end{aligned} \tag{3}$$

Our goal is to write $D^2 K(u, v)[(\hat{u}, \hat{v}), (\hat{u}, \hat{v})]$ as $(\hat{u}, \hat{v})^T \cdot A \cdot (\hat{u}, \hat{v})$ where the symmetric matrix A is the matrix associated with the quadratic form. We reorganize the equation by grouping $(\hat{a}_x, \hat{b}_y)$ together, where a, b is u or v , and x, y is 1 or 2. Recall that $\langle p, iq \rangle = \langle p_2 q_1 - p_1 q_2 \rangle$.

$$\begin{aligned}
D^2 K_{\mu,c}(u, v)[(\hat{u}, \hat{v}), (\hat{u}, \hat{v})] = & \\
& + (\hat{u}_1^2)(1) \\
& + (\hat{u}_2^2)(1) \\
& + (\hat{u}_1 \hat{v}_1)(-8v_1 v_2) \\
& + (\hat{u}_1 \hat{v}_2)(-4v_1^2 - 12v_2^2 - 2\mu) \\
& + (\hat{u}_2 \hat{v}_1)(12v_1^2 + 4v_2^2 - 2\mu) \\
& + (\hat{v}_1^2) \left(2c - 4u_1 v_2 + 12u_2 v_1 - \frac{48v_1^2 \mu |v|^2}{|2v^2 - 1|^5} (4|v|^4 - 4|v|^2 + 1) \right. \\
& + \frac{4\mu |v|^2}{|2v^2 - 1|^3} (4v_1^2 + 2|v|^2 - 1) + \frac{16v_1^2 \mu}{|2v^2 - 1|^3} (2|v|^2 - 1) - \frac{2\mu}{|2v^2 - 1|} \Big) \\
& + (\hat{u}_2 \hat{v}_2)(8v_1 v_2) \\
& + (\hat{v}_1 \hat{v}_2) \left(-8u_1 v_1 + 8u_2 v_2 - \frac{96v_1 v_2 \mu |v|^2}{|2v^2 - 1|^5} (4|v|^4 - 1) \right. \\
& + \frac{32v_1 v_2 \mu |v|^2}{|2v^2 - 1|^3} + \frac{64v_1 v_2 \mu |v|^2}{|2v^2 - 1|^3} \Big) \\
& + (\hat{v}_2^2) \left(2c - 12u_1 v_2 + 4u_2 v_1 - \frac{48v_2^2 \mu |v|^2}{|2v^2 - 1|^5} (4|v|^4 + 4|v|^2 + 1) \right. \\
& + \frac{4\mu |v|^2}{|2v^2 - 1|^3} (4v_2^2 + 2|v|^2 + 1) + \frac{16v_2^2 \mu}{|2v^2 - 1|^3} (2|v|^2 + 1) - \frac{2\mu}{|2v^2 - 1|} \Big)
\end{aligned} \tag{4}$$

Next, we construct the symmetric matrix A using equation (4), dividing the non-diagonal elements in half when necessary:

$$\begin{bmatrix}
1 & 0 & -4v_1 v_2 & -2v_1^2 - 6v_2^2 - \mu \\
0 & 1 & 6v_1^2 + 2v_2^2 - \mu & 4v_1 v_2 \\
-4v_1 v_2 & 6v_1^2 + 2v_2^2 - \mu & A_{33} & A_{43} \\
-2v_1^2 - 6v_2^2 - \mu & 4v_1 v_2 & A_{34} & A_{44}
\end{bmatrix}$$

$$A_{33} = 2c - 4u_1 v_2 + 12u_2 v_1 - \frac{48v_1^2 \mu |v|^2}{|2v^2 - 1|^5} (4|v|^4 - 4|v|^2 + 1) + \frac{4\mu |v|^2}{|2v^2 - 1|^3} (4v_1^2 + 2|v|^2 - 1) + \frac{16v_1^2 \mu}{|2v^2 - 1|^3} (2|v|^2 - 1) - \frac{2\mu}{|2v^2 - 1|}$$

$$A_{43} = A_{34} = -4u_1v_1 + 4u_2v_2 - \frac{48v_1v_2\mu|v|^2}{|2v^2-1|^5}(4|v|^4 - 1) + \frac{16v_1v_2\mu|v|^2}{|2v^2-1|^3} + \frac{32v_1v_2\mu|v|^2}{|2v^2-1|^3}$$

$$A_{44} = 2c - 12u_1v_2 + 4u_2v_1 - \frac{48v_2^2\mu|v|^2}{|2v^2-1|^5}(4|v|^4 + 4|v|^2 + 1) + \frac{4\mu|v|^2}{|2v^2-1|^3}(4v_2^2 + 2|v|^2 + 1) + \frac{16v_2^2\mu}{|2v^2-1|^3}(2|v|^2 + 1) - \frac{2\mu}{|2v^2-1|}$$

Lastly, the energy matrix K is formed: $K = [\hat{u}_1, \hat{u}_2, \hat{v}_1 \hat{v}_2] A [\hat{u}_1, \hat{u}_2, \hat{v}_1 \hat{v}_2]^T$.

5 Theory on the Existence of Global Surface of Section for Convex Hamiltonians

Our goal is see when the energy surface (expressed by $K^{-1}(0)$) is strictly convex, for the cases $\mu = 0, c = 1.601$ and $\mu = \frac{1}{2}, c = 2.01$. The strict convexity condition is $D^2K(u, v)(\hat{u}, \hat{v}) \geq a(\hat{u}, \hat{v})$ for some $a > 0$ and for all (u, v) and all $(\hat{u}, \hat{v})$. For this, it is sufficient (but not necessary) to show that matrix A is strictly positive definite. Strict positive definiteness can be checked using an application of the Sylvester criterion, by calculating where the determinants of the leading principal minors of A are greater than some positive constant $a > 0$ for all (u, v) . In the case of an intersection between the energy curve $K^{-1}(0)$ and $\det(A_i)=0$ (where A_i is one of the minors) in at least one of the slices, strong convexity fails and we would not be able to conclude the existence of a global surface of section.

The strict convexity of the energy implies that there is a disk-like global surface of section for the Hamiltonian flow[7]. A disk-like global surface of section is a 2-dimensional disk whose boundary is a periodic orbit, and whose interior is transverse to the flow, with the property that each trajectory (except for the periodic orbit) intersects the disk transversely (not tangentially) in both forward and backwards time. This allows us to define a Poincaré first return map that is well defined on the whole interior of the disk.

For example, let us examine the Poincaré section in figure 3.

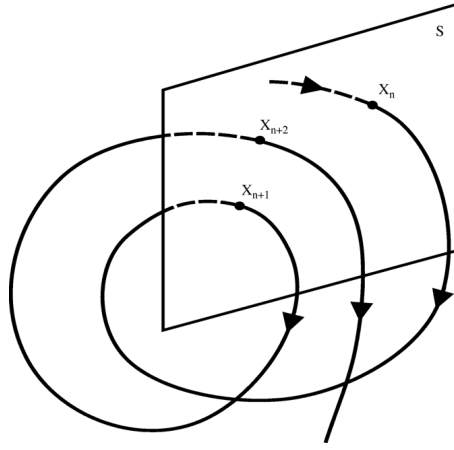


Figure 3: Poincaré section example (picture taken from Google images)

We have a function $f : S \rightarrow S$, where $S \subseteq \mathbb{R}^3$; of the three dimensional energy manifold $K^{-1}(0)$.

Define $f(x_i) = x_{i+1}$ where x_{i+1} is the first intersection point with S of the Hamiltonian flow ϕ^t with initial condition x_i . That is $f(x_i) = x_{i+1}$ if $x_{i+1} = \phi^{\tau(x_i)}(x_i)$ where $\tau(x_i) = \min\{t > 0 : \phi^t(x_i) \in S\}$. Observe that every input has an output, there are no tangential intersections, and no orbits escape. A such, a global Poincaré surface captures all the information about the dynamics of the flow.

Wherever the energy manifold K is positive definite, it is strictly convex, and the 3-dimensional dynamics of the flow on the energy manifold can be reduced to 2-dimensional dynamics of the Poincare map on the disk. We will use this tool for dimensionality reduction in our analysis. Since in our case the Hamiltonian flow is symplectic, it follows that the Poincaré map is area preserving. As a consequence of the Brouwer translation theorem, it follows that f has a fixed point in the disk. Consequently, the flow has a periodic orbit.

6 Graphs

Using calculations from previous sections, we investigate several different cases:

- $\mu = 0$ and $-c$ close to $-\frac{3}{2} = -1.5$, which is the critical energy $H(L_1)$. In particular, we examine the cases for $c = 1.601$. Note that for $c = 1.601$, $-c = -1.601$ is below $H(L_1) = -1.5$ as μ approaches 0 or 1.

- $\mu = \frac{1}{2}$ and $-c$ close to $H(L_1)$. We continue our analysis for $-c$ at a value slightly less than -2 .

We plot the graph where $u_1 = v_2 = \mu = 0, c = 1.601$ using matlab.

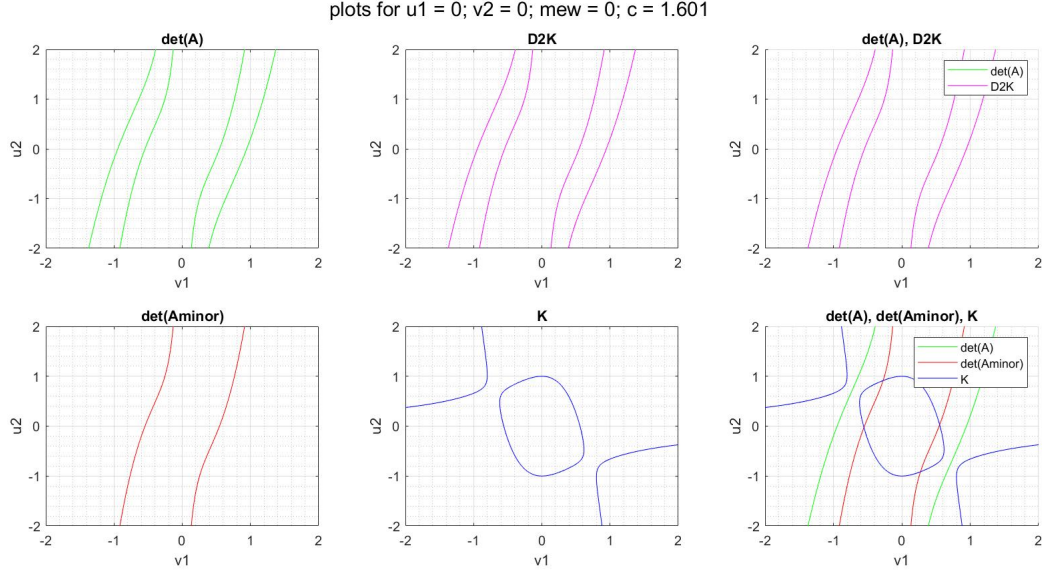


Figure 4: plots for $u_1 = v_2 = \mu = 0$

The top row indicates that the determinant of matrix A is equal to $\det(D^2K)$, as the green and purple graphs overlap perfectly. The bottom row shows that the determinant of $Aminor$, the top left 3×3 elements in matrix A , overlaps perfectly with the middle two lines of $\det(A)$.

Next, we examine where this slice of matrix A is positive definite. We accomplish this graphing the region where the determinants of the 4×4 matrix A and 3×3 matrix $Aminor$ are positive (using Mathematica). Note that the other two minors have a positive determinant everywhere.

The green and red curves in the figures represent the points where $\det(A)$ and $\det(Aminor)$ are equal to zero. On one side of these curves $\det(A)$ or $\det(Aminor)$ is positive and on the other side it is negative.

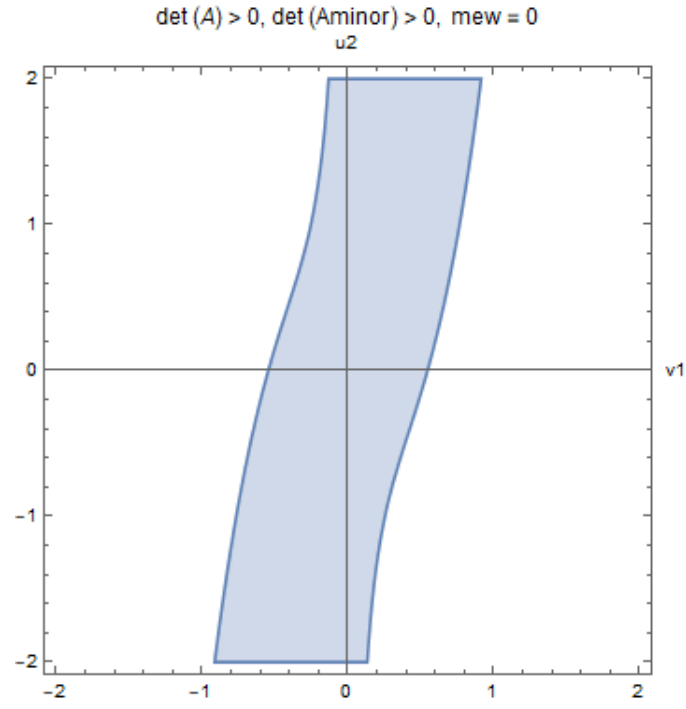


Figure 5: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_1 = v_2 = \mu = 0$

We decrease the energy (by increasing c), and check where the energy curve K is tangent but not crossing $\det(A)$ or $\det(A_{\text{minor}})$.

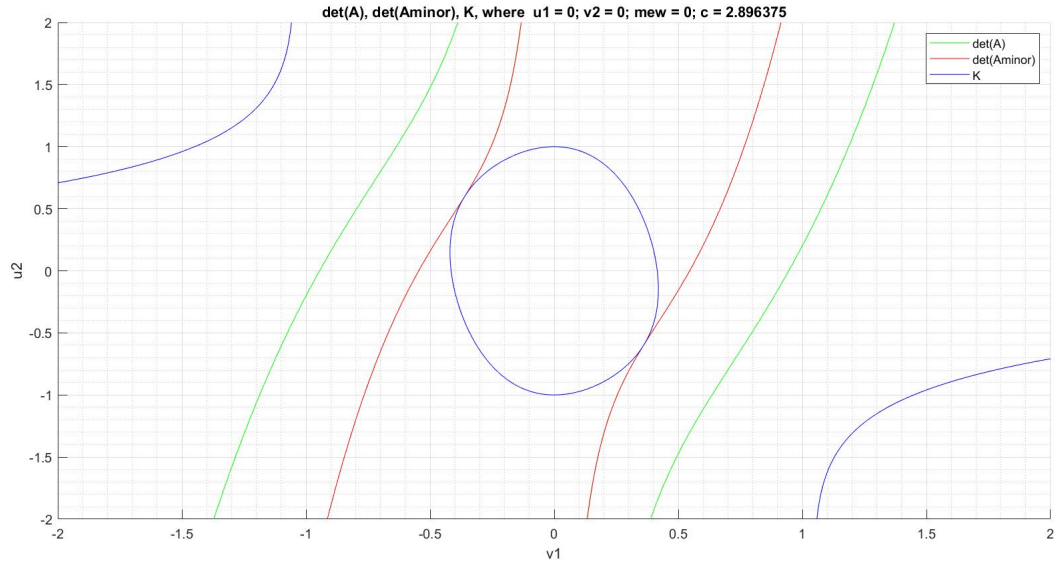


Figure 6: plot for $u_1 = v_2 = \mu = 0$, $c = 2.896375$

The numerical evidence for the case $c \geq 2.896375$, where $u_1 = v_2 = \mu = 0$ indicates that strong convexity holds, in which case we can conclude the existence of a global surface of section.

We now examine other slices for $\mu = 0$, $c = 1.601$.

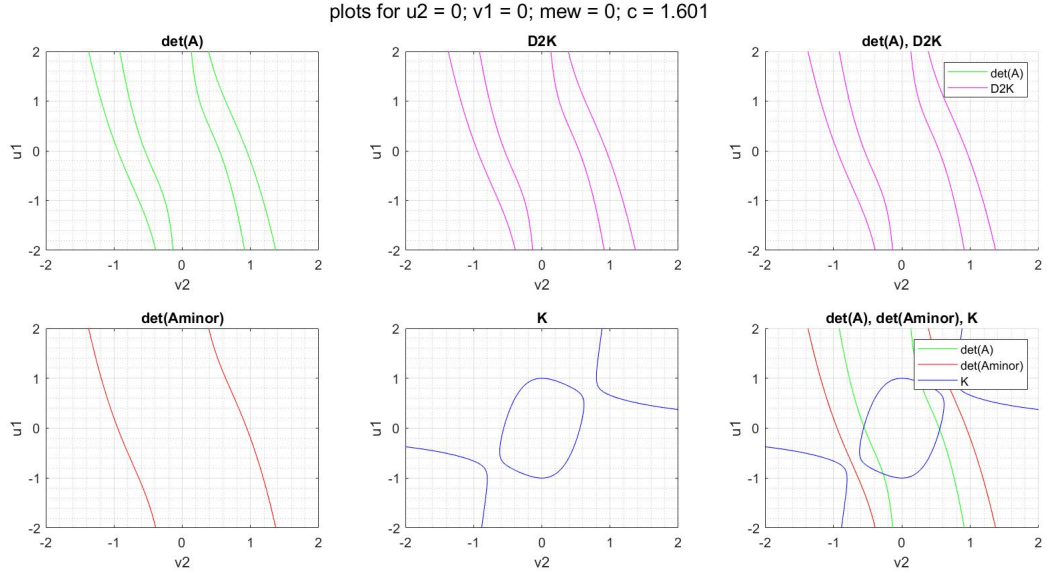


Figure 7: plots for $u_2 = v_1 = \mu = 0$

The plots for $u_2 = v_1 = 0$ look like the mirror image of figure 4 (plots for $u_1 = v_2 = 0$). In this case, $\det(\text{Aminor})$ overlaps with the outer two lines instead of the inner two lines.

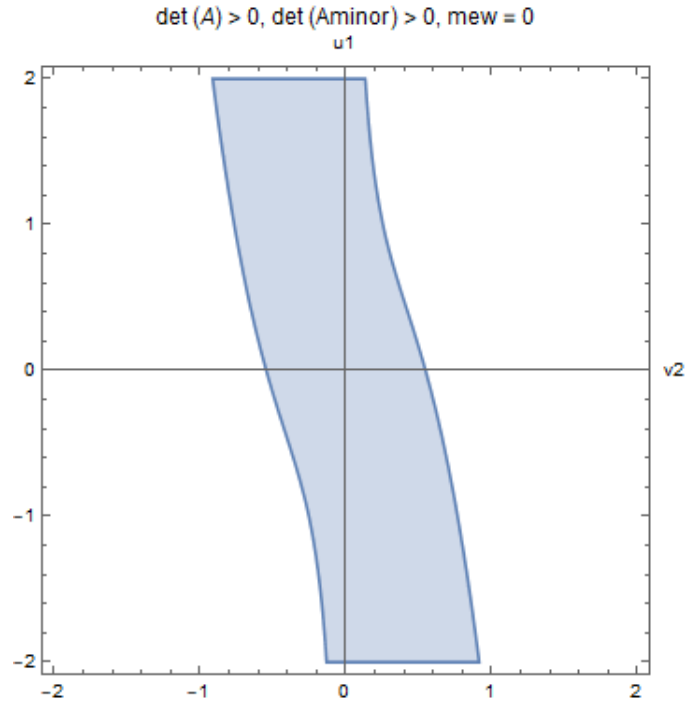


Figure 8: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_2 = v_1 = \mu = 0$

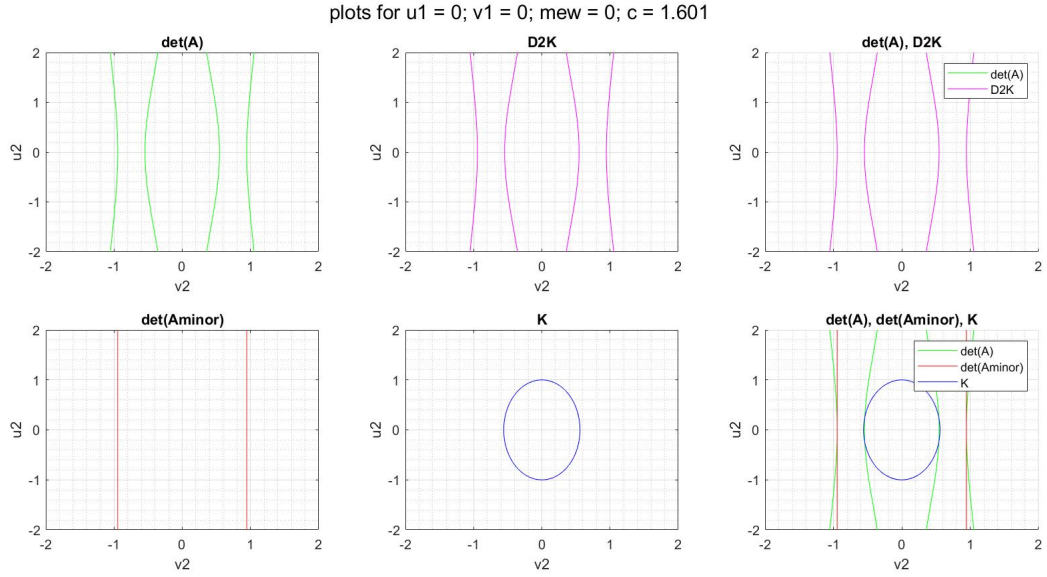


Figure 9: plots for $u_1 = v_1 = \mu = 0$

We see in the $u_1 = v_1 = 0$ case that $\det(A)$ is equal to $\det(D^2K)$, and the determinant of A_{minor} touches $\det(A)$ but does not completely overlap it.

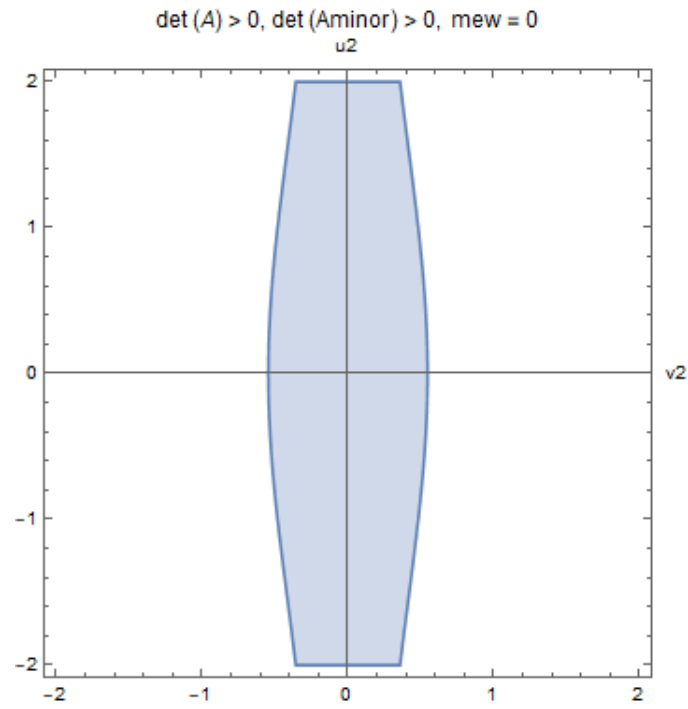


Figure 10: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_1 = v_1 = \mu = 0$

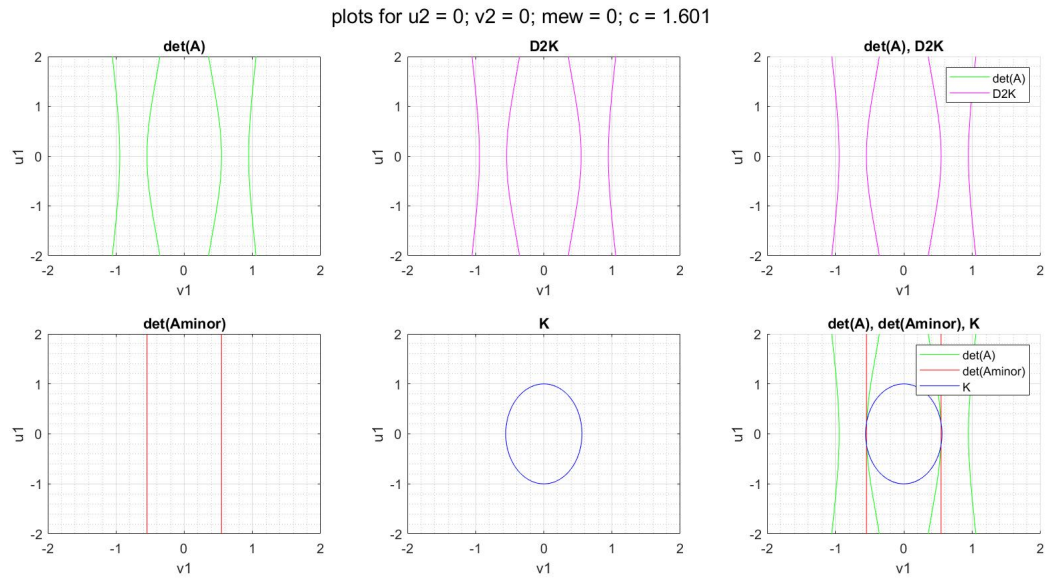


Figure 11: plots for $u_2 = v_2 = \mu = 0$

The plots for $u_2 = v_2 = 0$ appear quite similar to figure 9 (plots for $u_1 = v_1 = 0$), except that $\det(A_{\text{minor}})$ touches

the inner two lines of $\det(A)$ instead of the outer two lines.

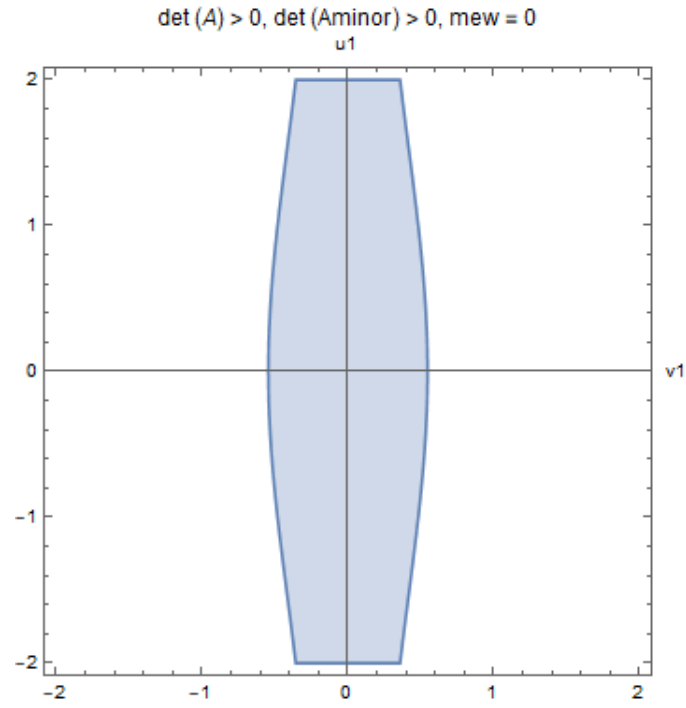


Figure 12: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_2 = v_2 = \mu = 0$

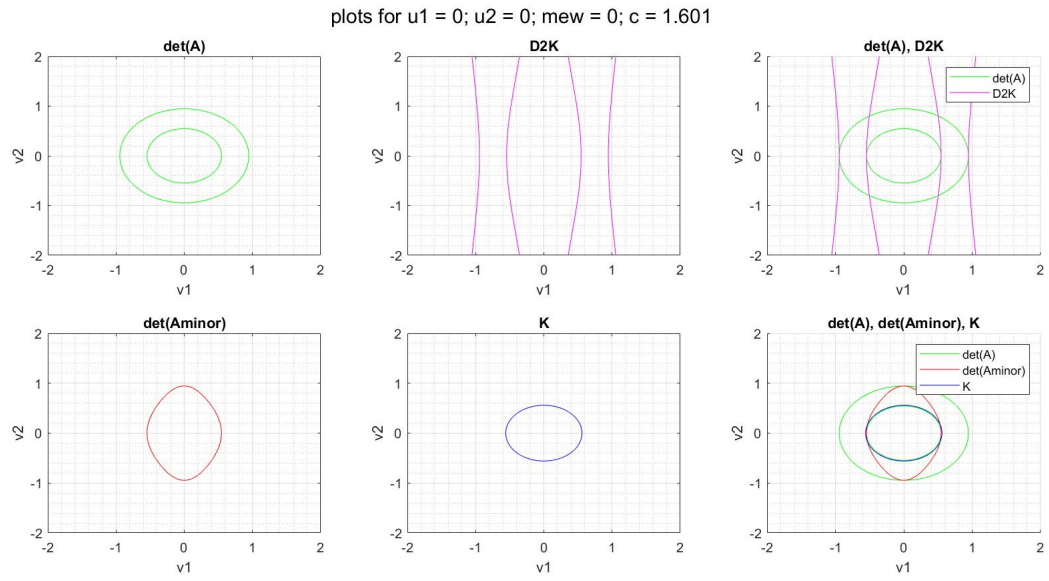


Figure 13: plots for $u_1 = u_2 = \mu = 0$

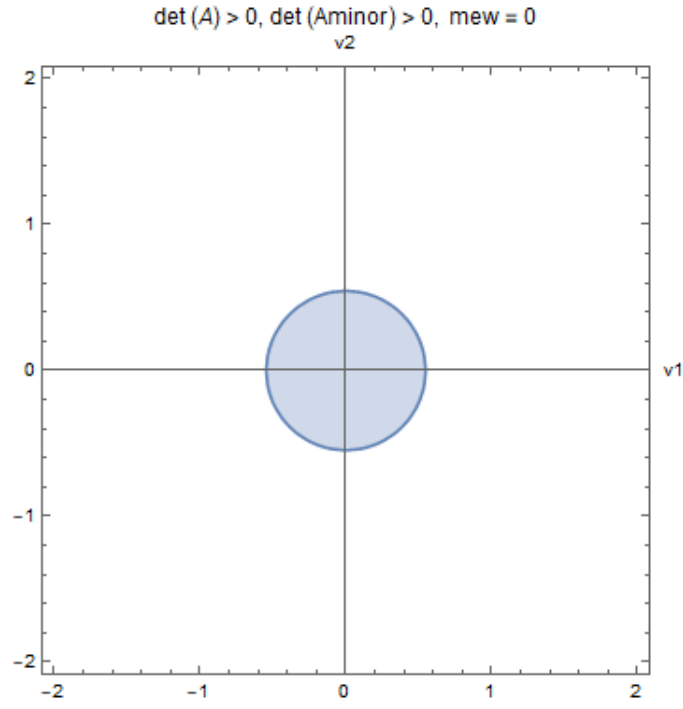


Figure 14: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_1 = u_2 = \mu = 0$

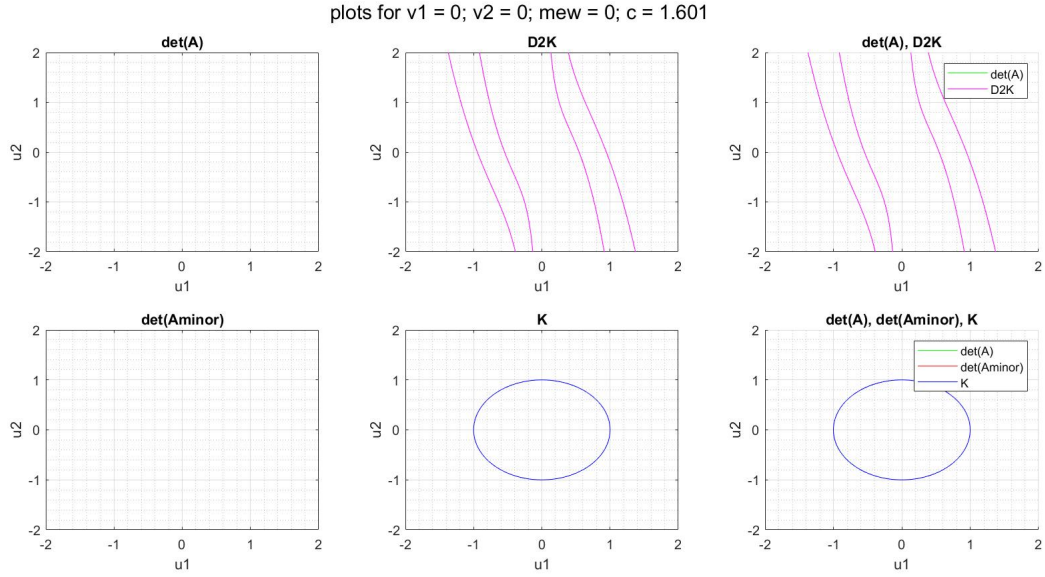


Figure 15: plots for $v_1 = v_2 = \mu = 0$

In the $v_1 = v_2 = \mu = 0$ case, we have $\det(A) = 2563201/250000$, and $\det(A_{\text{minor}}) = 1601/500$, both determinants are positive everywhere.

In conclusion, K is not strictly convex for the values $\mu = 0$, $c = 1.601$ because the curves K and $\det(A)$, or K and $\det(A_{\text{minor}})$, cross each other in some slices. We cannot conclude the existence of a global surface of section for these values.

Next, we examine the cases where $\mu = 0.5$. Note that the previous equation for $\det(D^2K)$ was only relevant for the case $\mu = 0$, therefore it is not necessary to include that curve for $\mu = 0.5$.

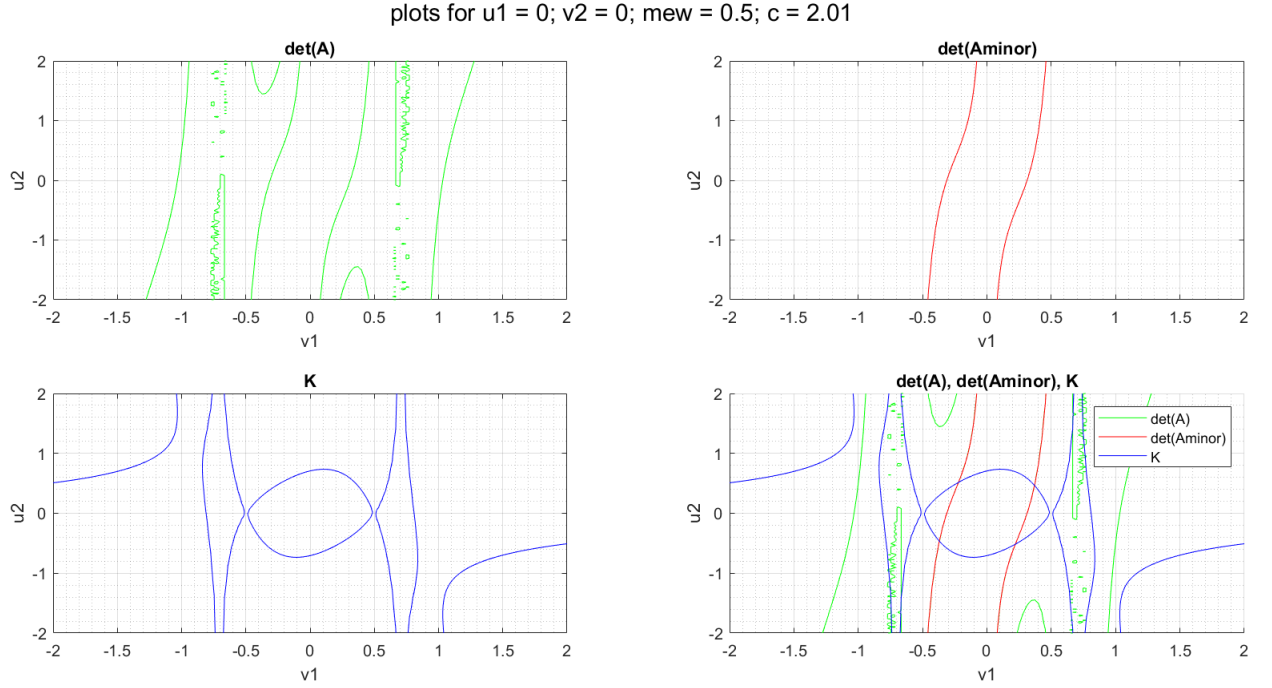


Figure 16: plots for $u_1 = v_2 = 0$, $\mu = 0.5$

As in the $\mu = 0$ case, the determinant of A_{minor} overlaps the determinant of A on the inner lines of $\det(A)$.

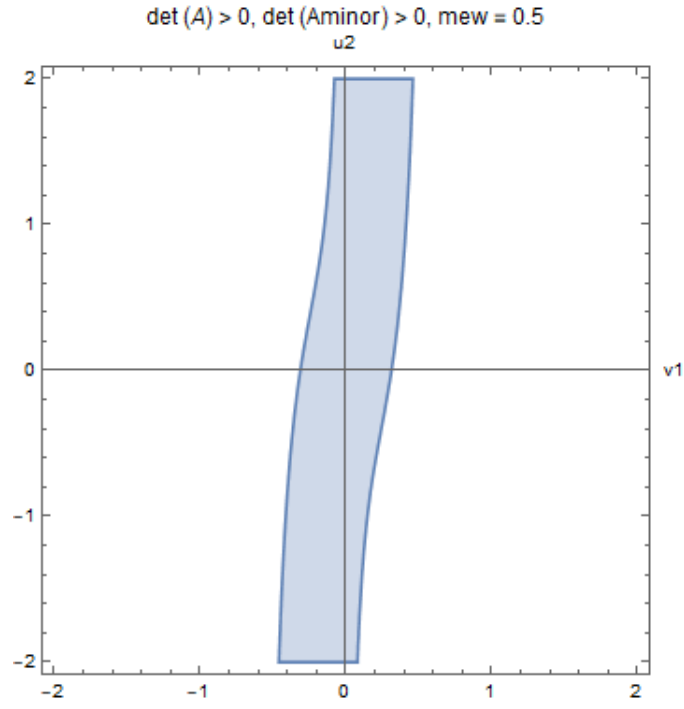


Figure 17: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_1 = v_2 = 0, \mu = 0.5$

We decrease the energy $-c$ (by increasing c), and check where the energy curve K is tangent but not crossing $\det(A)$ or $\det(A_{\text{minor}})$.

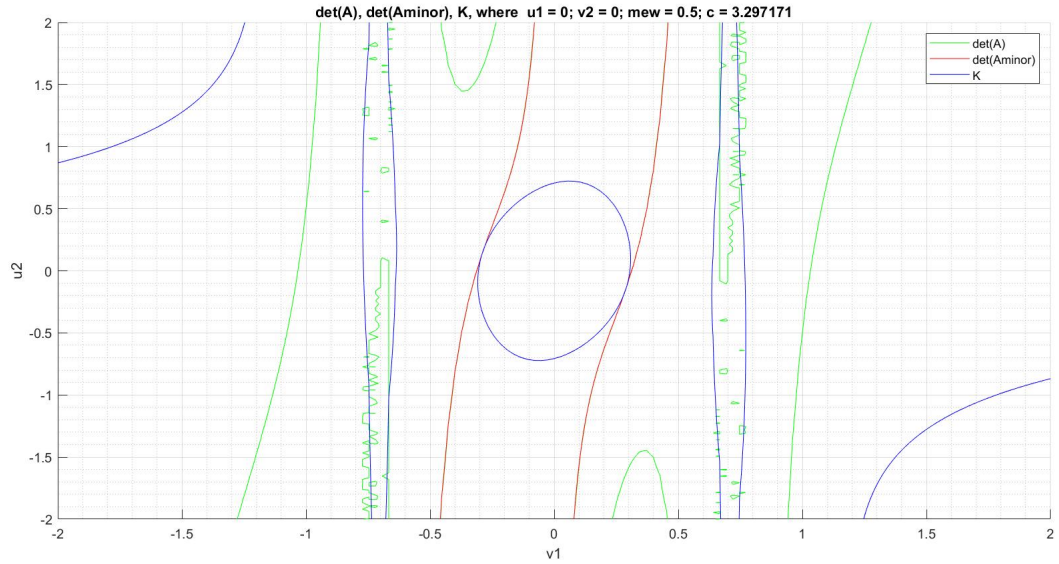


Figure 18: plot for $u_1 = v_2 = \mu = 0, c = 3.297171$

The numerical evidence for the case $c \geq 3.297171$, $u_1 = v_2 = 0, \mu = 0.5$ indicates that strong convexity holds.

We can conclude the existence of a global surface of section for these values.

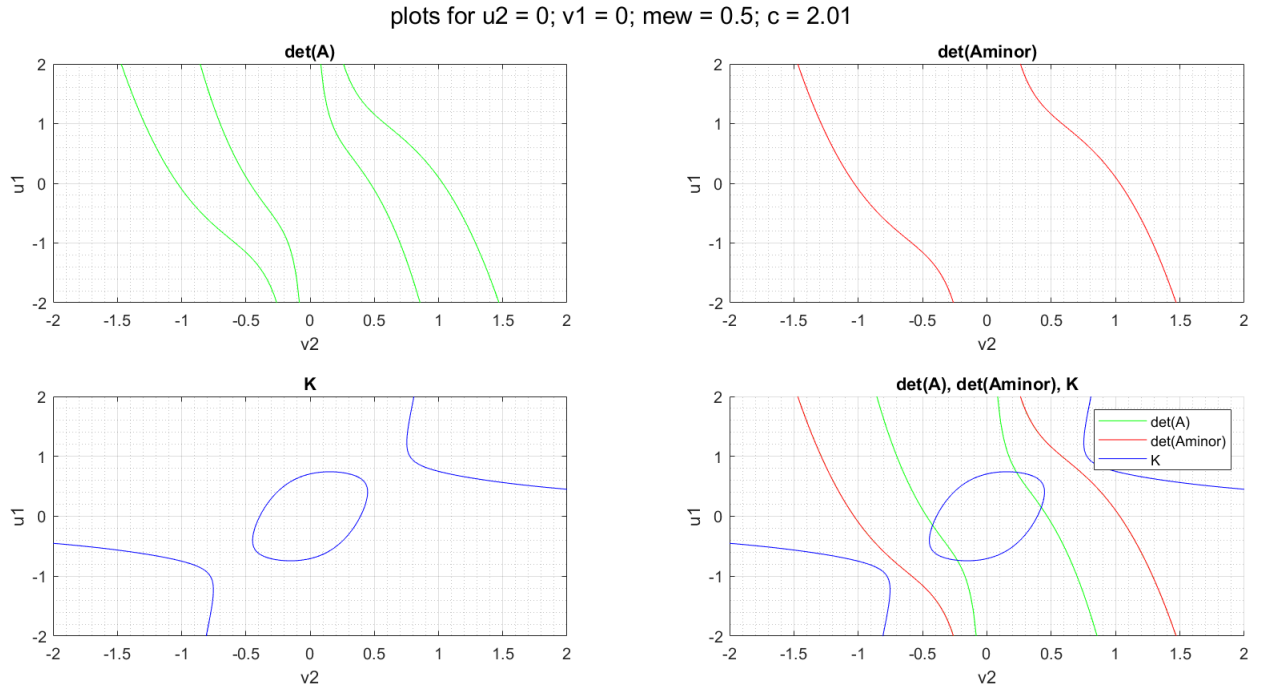


Figure 19: plots for $u_2 = v_1 = 0, \mu = 0.5$

As in the $\mu = 0$ case, the determinant of Aminor overlaps the determinant of A on the outer lines of det(A).

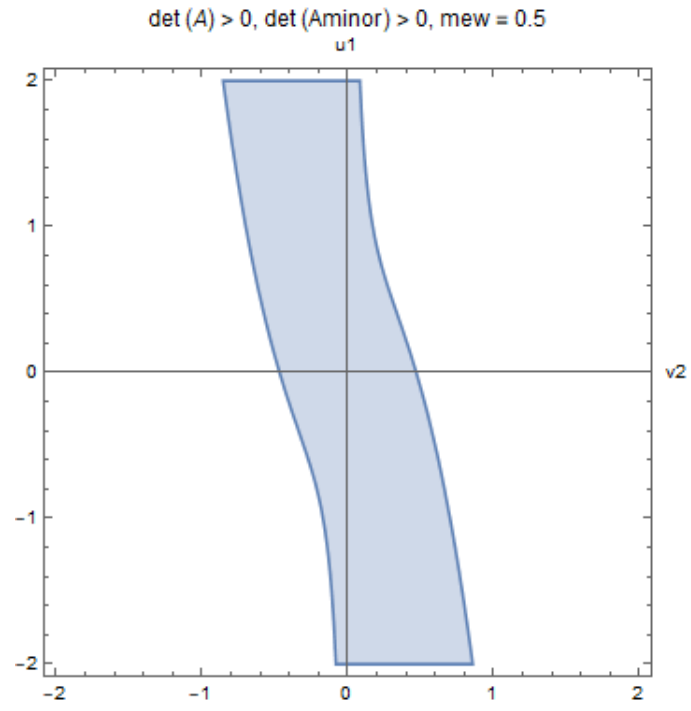


Figure 20: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_2 = v_1 = 0, \mu = 0.5$

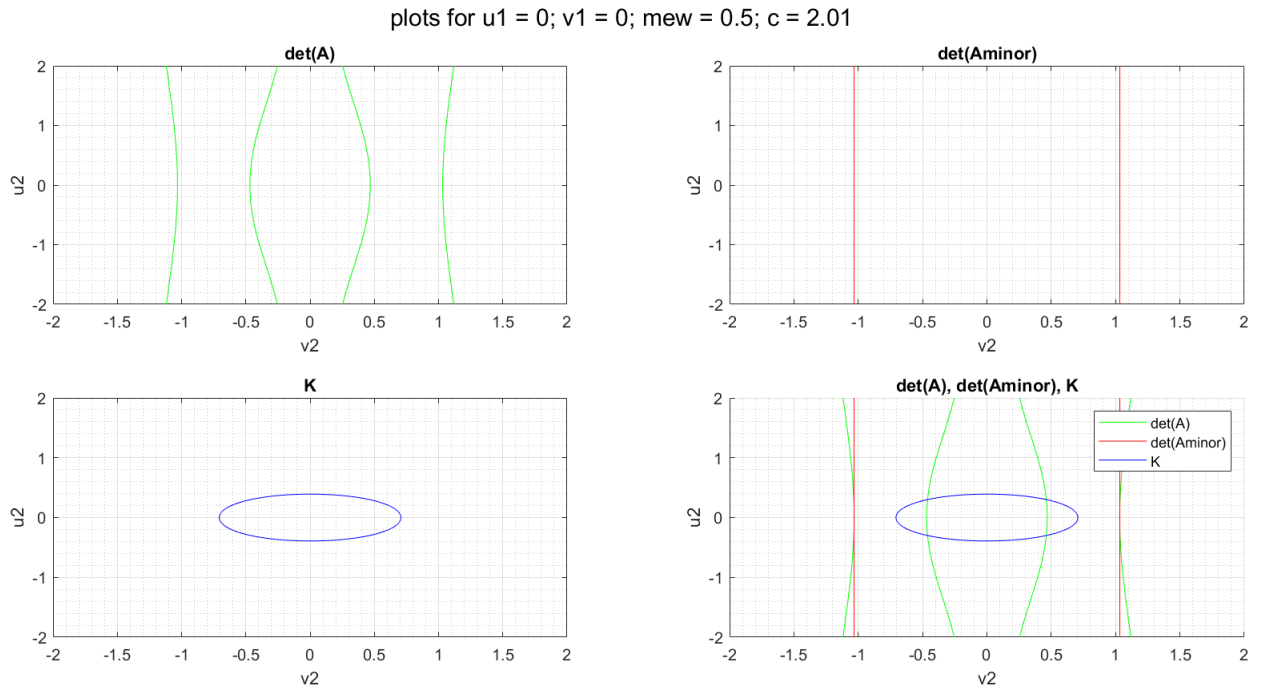


Figure 21: plots for $u_1 = v_1 = 0, \mu = 0.5$

As in the $\mu = 0$ case, the determinant of A_{minor} touches the determinant of A on the outer lines but does not overlap it.

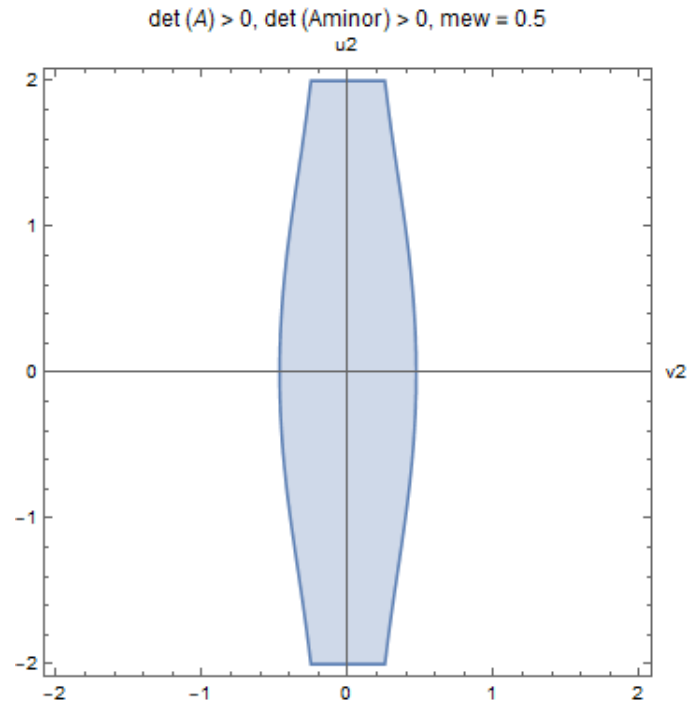


Figure 22: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_1 = v_1 = 0, \mu = 0.5$

plots for $u_2 = 0; v_2 = 0; \mu = 0.5; c = 2.01$

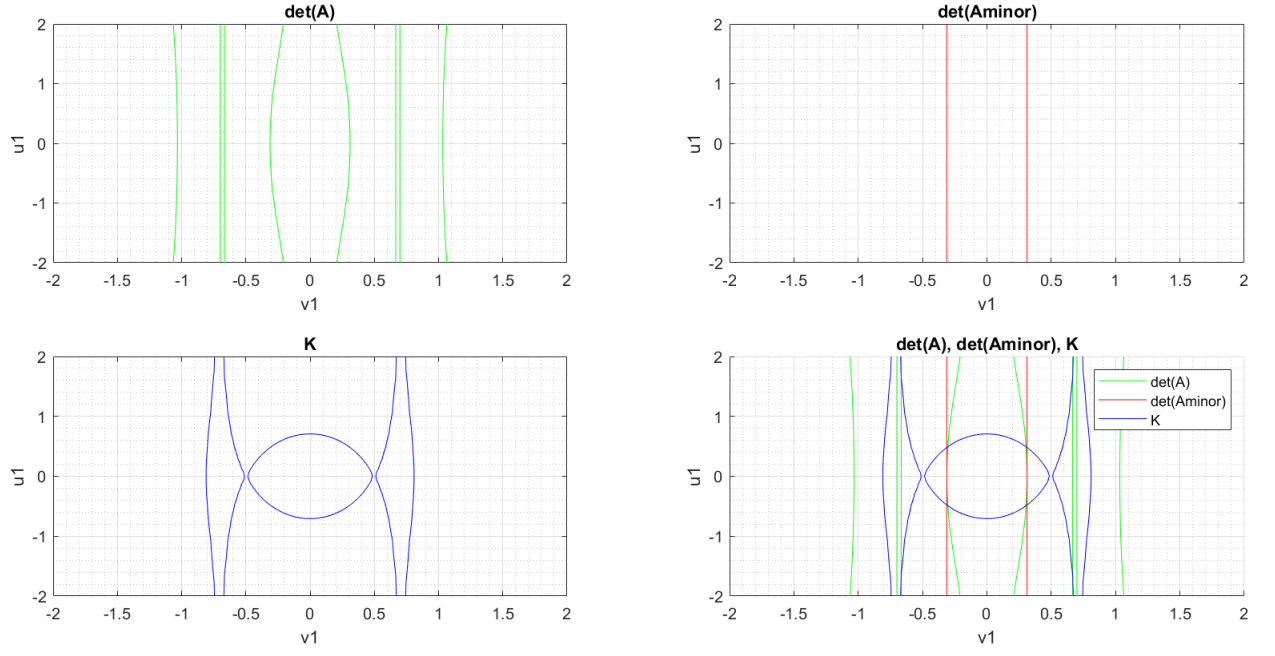


Figure 23: plots for $u_2 = v_2 = 0, \mu = 0.5$

As in the $\mu = 0$ case, the determinant of A_{minor} touches the determinant of A on the inner lines but does not overlap it.

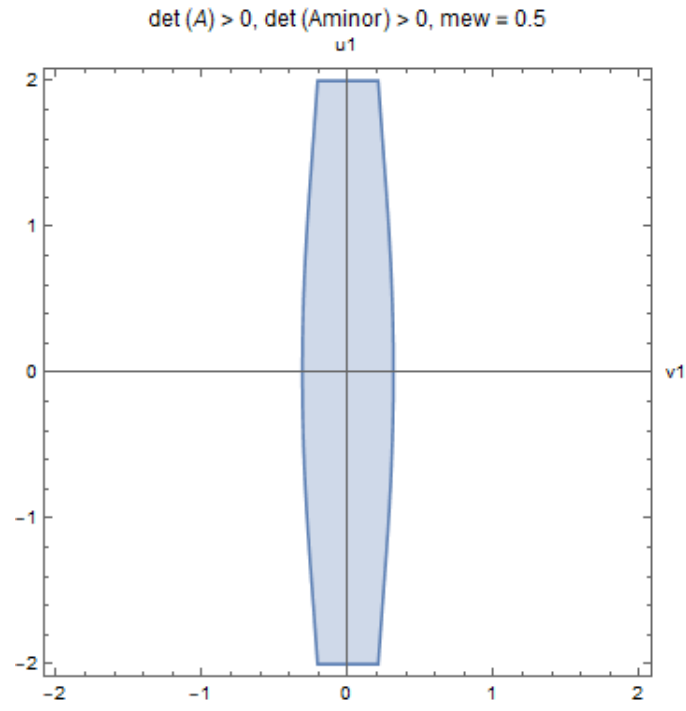


Figure 24: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_2 = v_2 = 0, \mu = 0.5$

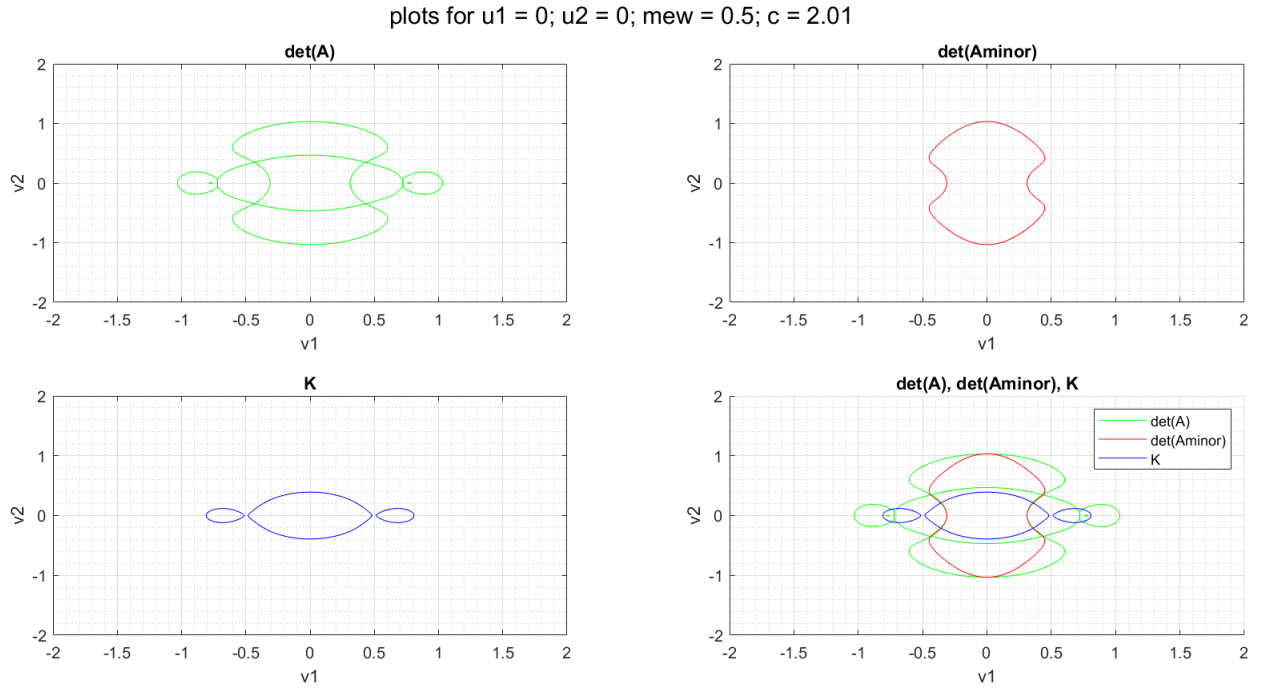


Figure 25: plots for $u_1 = u_2 = 0, \mu = 0.5$

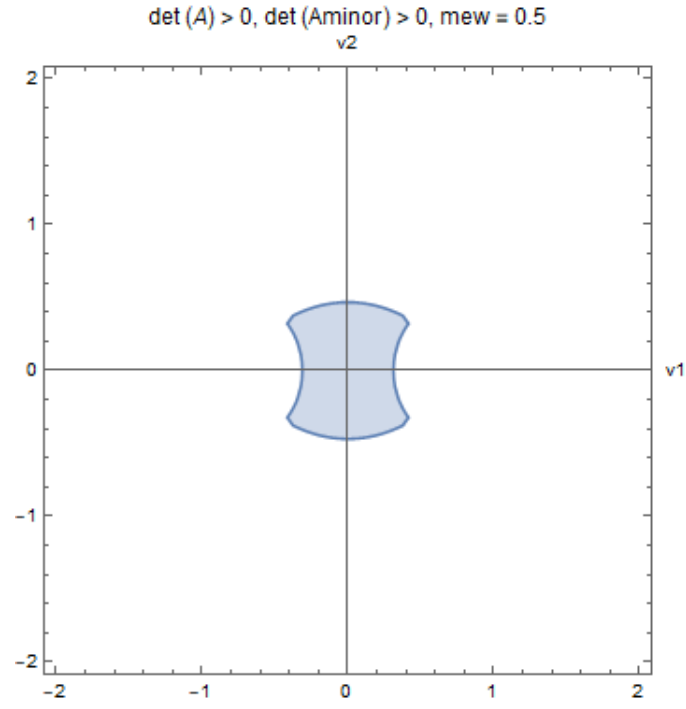


Figure 26: region where $\det(A)$ and $\det(A_{\text{minor}})$ are positive for $u_1 = u_2 = 0, \mu = 0.5$

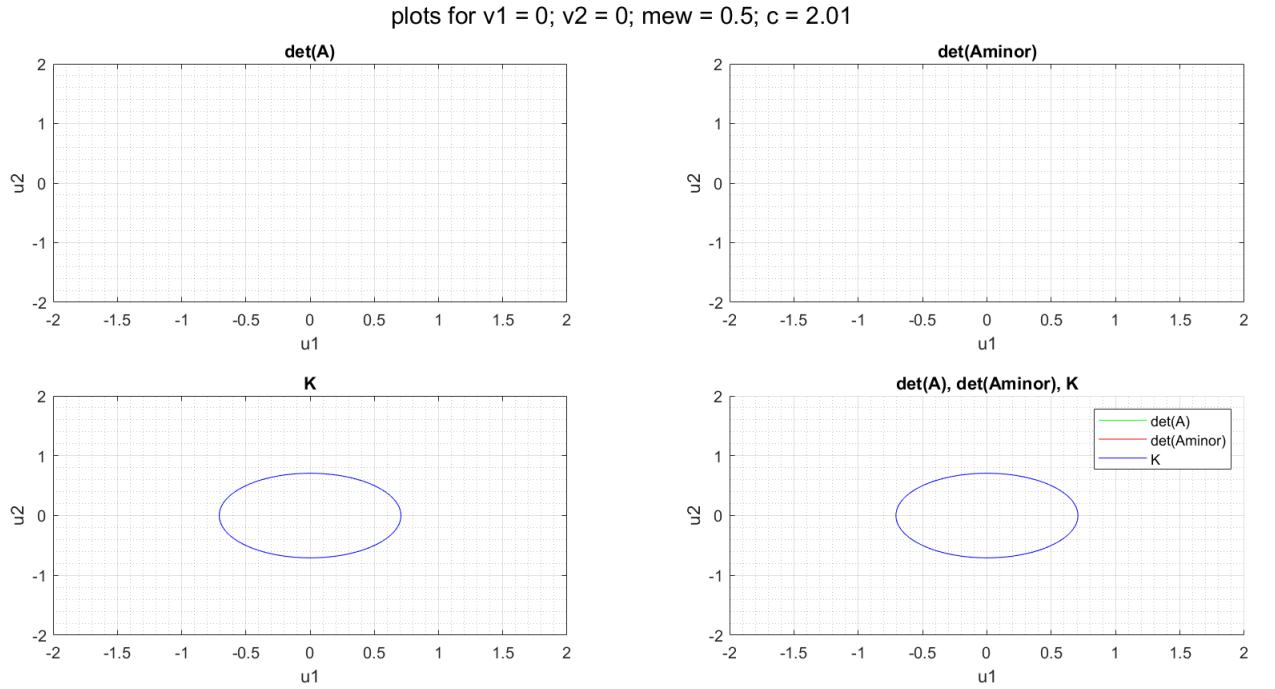


Figure 27: plots for $v_1 = v_2 = 0, \mu = 0.5$

In the $v_1 = v_2 = 0, \mu = 0.5$ case, $\det(A) = 59536/15625$, and $\det(A_{\text{minor}}) = 244/125$: both determinants are positive everywhere.

In conclusion, K is not strictly convex for the values $\mu = 0.5, c = 2.01$ because the curves K and $\det(A)$, or K and $\det(A_{\text{minor}})$, cross each other in some slices. We cannot conclude the existence of a global surface of section for these values.

7 Conclusions

In this paper, we analyzed the restricted planar three body problem as it applies to the Sun, Earth, and a spacecraft. We derived the locations of the first and second Lagrangian equilibrium points using two methods. Next, we set up our Hamiltonian and Hessian, constructing matrix A which helped us form the energy matrix. Whenever matrix A is strictly positive definite, the energy surface is convex and there exists a disk like global surface of section for the Hamiltonian flow. To see where this condition applies, we plotted the function K , along with the determinants of A and A_{minor} , and found the positive definite regions on all the slices.

Future directions of this research include using advanced methods to decide convexity, and numerically finding a precise value of c_0 for specific μ, u, v such that for $c > c_0$ strong convexity holds, which will indicate the existence of a global surface of section for these values.

Acknowledgements

I would like to express my deep, sincere gratitude to Professor Marian Gidea, my advisor, for his invaluable guidance and support. The work for this project spanned various locations throughout "normal" times as well as the present day pandemic. Nevertheless, Professor Gidea's door (both virtual and real) has remained wide open for questions, advice, and encouragement at every juncture. In his capacity as both mentor and professor through the duration of my BA and MA studies at Yeshiva University, Professor Gidea has taught me more than I could ever give him credit for here, and always pushed me to reach my potential. Through his example, I have come to understand and appreciate the beauty of mathematics as well as the art of its teaching. By emulating Professor Gidea, whose expertise in mathematics is complemented by his gift for pedagogy, I have begun to develop my own skills as educator, researcher, and mentor. He has truly inspired me, along with so many others.

Likewise, I am indebted to Professor Morton Lowengrub, who opened the door to higher mathematics for his students and invited me in with open arms. By establishing the Yeshiva University BA-MA program, Professor Lowengrub afforded me a taste of high-level mathematics during my undergraduate years which left me craving for more. He spent countless hours patiently explaining mathematical concepts to me, while providing encouragement, valuable advice, and expert guidance. By setting high expectations early on, while I was yet an undergraduate, Professor Lowengrub impelled me never to settle for less than my best work, and for this I will be eternally grateful.

I would be remiss if I did not express my sincere appreciation to Professor Mina Teicher of Bar Ilan University. Professor Teicher invited me to join her lab as an undergraduate, even though I had no prior research experience. It was she who first exposed me to the exciting world of scientific and mathematical research and has continued to serve as an unfailing resource for advice and support. I have since traveled to various conferences and other events with Professor Teicher, experiences that have provided me with a plethora of opportunities to learn about the world of higher academia and make the acquaintance of senior mathematicians. Professor Mina Teicher has been a true role model: as mathematician, mentor, and human being. She has held my hand through every step of the way and

I hope she never lets go.

Last, but certainly not least, I am forever grateful to my family and friends who provided me with unfailing support and encouragement throughout my studies. They have kept me going, and this project would not have been possible without them.

References

- [1] Frauenfelder, Urs, and Otto Van Koert. *The Restricted Three Body Problem and Holomorphic Curves*. Springer International Publishing, 2018.
- [2] Offin, Dan, and R. Glen. Hall, and KR Meyer, Introduction to Hamiltonian Dynamical Systems and the n-body Problem.” *Applied Mathematical Sciences* 90.
- [3] Szebehely, Victor. *Theory of Orbits: the Restricted Problem of Three Bodies*. Yale University New Haven CT, 1967.
- [4] Belbruno, Edward. *Capture Dynamics and Chaotic Motions in Celestial Mechanics: With Applications to the Construction of Low Energy Transfers*. Princeton University Press, 2018.
- [5] James Stewart, ”Single Variable Essential Calculus”, Thomson Brooks/Cole, 2012, p. 189
- [6] Albers, P., Fish, J. W., Frauenfelder, U., Hofer, H., and Van Koert, O ”Global Surfaces of Section in the Planar Restricted 3-Body Problem.” *Archive for Rational Mechanics and Analysis*, vol. 204, no. 1, 2011, pp. 273-284
- [7] Hofer, H., Wysocki, K., Zehnder, E. (1998). The Dynamics on Three-Dimensional Strictly Convex Energy Surfaces. *Annals of Mathematics*, 148(1), second series, 197-289. doi:10.2307/120994